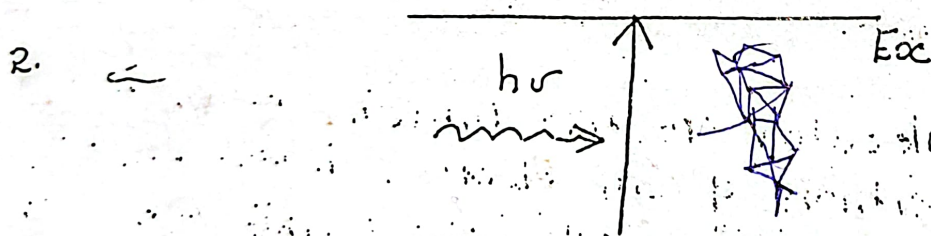


Spectroscopy is the branch of science, which deals with the study of the interaction of electromagnetic radiation with matter. Spectroscopy is one of the most advanced techniques available for the determination of the structure of atoms and molecules.

Types of spectra

Based on the nature of interaction, the spectra can be broadly divided into the following types

1. Absorption spectrum: results when an atom or molecule undergoes a transition from lower energy level to the higher energy with the absorption of a photon of energy $h\nu$.

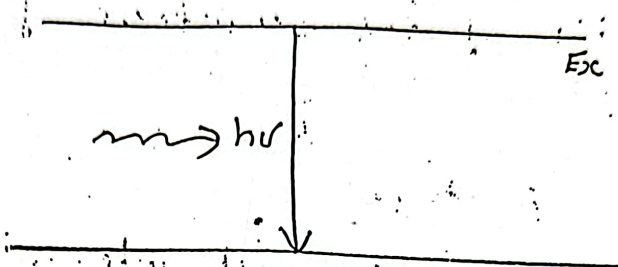


3. Emission spectrum

The important special techniques such as UV-visible, IR, Microwave, and NMR belong to absorption spectra.

2. Emission spectra: When an atom or molecule undergoes a transition from higher energy level (E_x) to lower energy level with an emission

of a photon of energy $h\nu$, the spectrum obtained is called an emission spectrum.
eg. Hydrogen spectrum, Atomic emission spectrum [AES]



Emission Spectrum. Egr

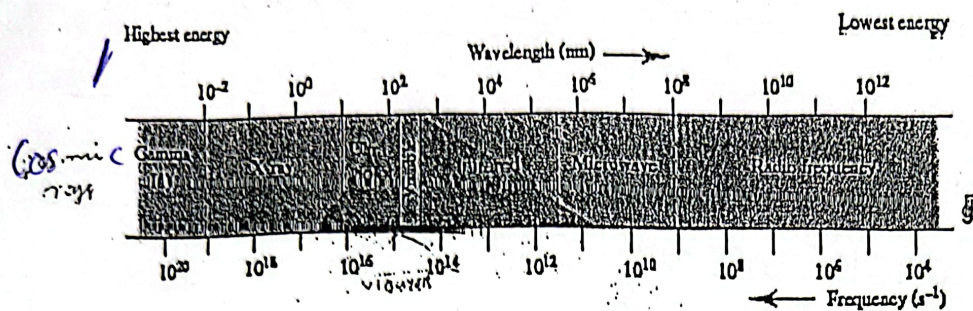
Based on the nature of the interacting species, the spectra can be divided into two categories

- ① Atomic Spectra: which arises from the transition of electrons between the atomic energy levels.
- ② Molecular Spectra: which arises from the transitions of an electron between the molecular energy levels. UV-visible, IR, Microwave and NMR spectra belongs to this category.

Electromagnetic spectrum:

The arrangement of different types of electromagnetic radiations in order of increasing wavelength (or decreasing frequency)

is known as electromagnetic spectrum.
The electromagnetic spectrum is shown in Figure.



1.1 Electromagnetic Spectrum

Visible spectrum has seven colours in order of increasing wavelength as VIBGYOR.

The velocity of light (c), frequency (ν) and wavelength (λ) are related by the equation

$$\nu = \frac{c}{\lambda}$$

Electromagnetic radiation has the properties of both particles and waves. The particles are called photons and each possesses an

(4)
amount of energy, referred to as a quantum.
According to Planck's quantum theory, the
energy of a photon

$$E = h\nu = h \frac{c}{\lambda}$$

$$E = hc\bar{\nu}$$

$$\nu = \frac{c}{\lambda}$$

$$\frac{1}{\lambda} = \bar{\nu}$$

h = Planck's constant and has the
value $h = 6.63 \times 10^{-34}$ J.s.

Molecular Energy levels:

Types of energy present in a molecule

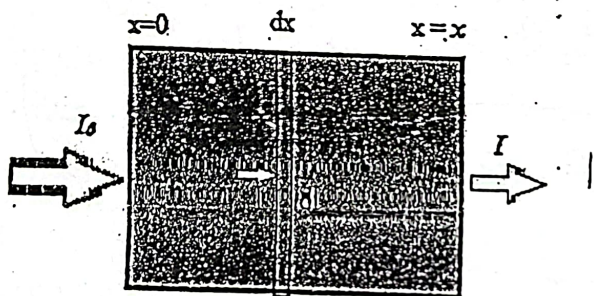
1. Translational Energy [E_{trans}]: is concerned with the overall movements of the molecules along three axes.
2. Rotational energy:- involves the rotational motion of the molecule.
3. Vibrational energy:- associated with vibrations within a molecule stretching and bending vibrations.
4. Electronic energy:- involves changes in the distribution of e⁻s by the promotion of e⁻s to higher levels on absorption of energy.

Therefore, the total energy of the molecule

$$E_{total} = E_{elec} + E_{vib} + E_{rot} \quad (\text{translation energy is negligibly small})$$

Law of spectroscopy

✓ Beers-Lambert's Law:- When a parallel beam of monochromatic electromagnetic radiation is passed through an absorbing solution of given concentration (c), the rate of decrease in intensity ($-dI$) of radiation with thickness of the solution (dx) is proportional to the intensity of incident radiation (I) at that point and also to the concentration (c) of the solution.



$$-\frac{dI}{dx} \propto I \cdot c$$

$$-\frac{dI}{I} = k \cdot c \cdot dx$$

Let I_0 be the intensity of incident radiation and I the intensity after passing through any finite thickness x of the medium.

$$-\int_{I_0}^I \frac{dI}{I} = k \cdot c \int_0^x dx \quad \left[\begin{array}{l} \text{Integrate with} \\ \text{limits} \end{array} \right]$$

$$-\left[\ln I \right]_{I_0}^I = k \cdot c [x]_0^x$$

(b)

$$-\left[\ln I - \ln I_0\right] = k c x$$

$$\ln \frac{I_0}{I} = k c x$$

$$2.303 \log \frac{I_0}{I} = k c x$$

$$\log \frac{I_0}{I} = \frac{k}{2.303} c x$$

$$\boxed{\log \frac{I_0}{I} = \epsilon c x \quad \left[\frac{k}{2.303} = \epsilon \right]}$$

Where $\epsilon = \frac{k}{2.303}$ called molar extinction

Coefficient, value of which depends on the nature of the absorbing molecule and type of absorption.

Logarithm of ratio of intensities of incident light (I_0) to the transmitted light (I) is called absorbance A

$$\log \frac{I_0}{I} = A \quad \text{--- (2)}$$

Comparing equ (1) and (2)

$$\boxed{\log \frac{I_0}{I} = A = \epsilon c x}$$

transmittance (T) is defined as the ratio of intensities of transmitted to incident light

(7)

transmittance $T = \frac{I}{I_0}$ where I_0 is incident light and I is transmitted light. $A = -\log \frac{I}{I_0}$

$A = -\log T$

UV-Visible Spectroscopy

[Electronic Spectroscopy]

UV-Region 200 - 400 nm

Visible Region 400 - 800 nm

UV-Visible Spectroscopy arises from the transition of electrons within a molecule or ion from a lower to higher energy level of the molecule. When a molecule absorbs UV or visible light, its electrons get promoted from ground state to the higher energy state, the energy difference between the two states is given by $\Delta E = h\nu$

where 'h' is Planck's constant

Also $\Delta E = E_2 - E_1$, E_1 = Energy of ground state

E_2 = Energy of excited state

we know that the total energy of a molecule is equal to the sum of the electronic, vibrational and rotational energies.

$E_{total} = E_{electronic} + E_{vibrational} + E_{rotational}$

total energy of a molecule $\approx E_{\text{elec}} + E_{\text{vib}} + E_{\text{rot}}$

The absorption in the UV-visible region involves single electronic transition from lower energy state to higher energy state, so the spectrum must contain a single discrete line. But in actual practice, a broad absorption band appears. This is because the electronic transitions are also accompanied by vibrational and rotational transitions. As the energy required for these transitions is much less so the broad band appears.

Electronic Transitions

When a molecule absorbs energy in the UV or visible region, its e⁻s are promoted from a bonding molecular orbital to an antibonding molecular orbital. The electronic transitions involved in the UV and visible region are of the following types.

(i) $\sigma \rightarrow \sigma^*$ transitions

(ii) $n \rightarrow \sigma^*$ transitions

(iii) $n \rightarrow \pi^*$ transitions

(iv) $\pi \rightarrow \pi^*$ transitions

Various electronic energy levels

$\sigma \rightarrow \sigma$ bonding

$\sigma^* \rightarrow \sigma$ anti bonding

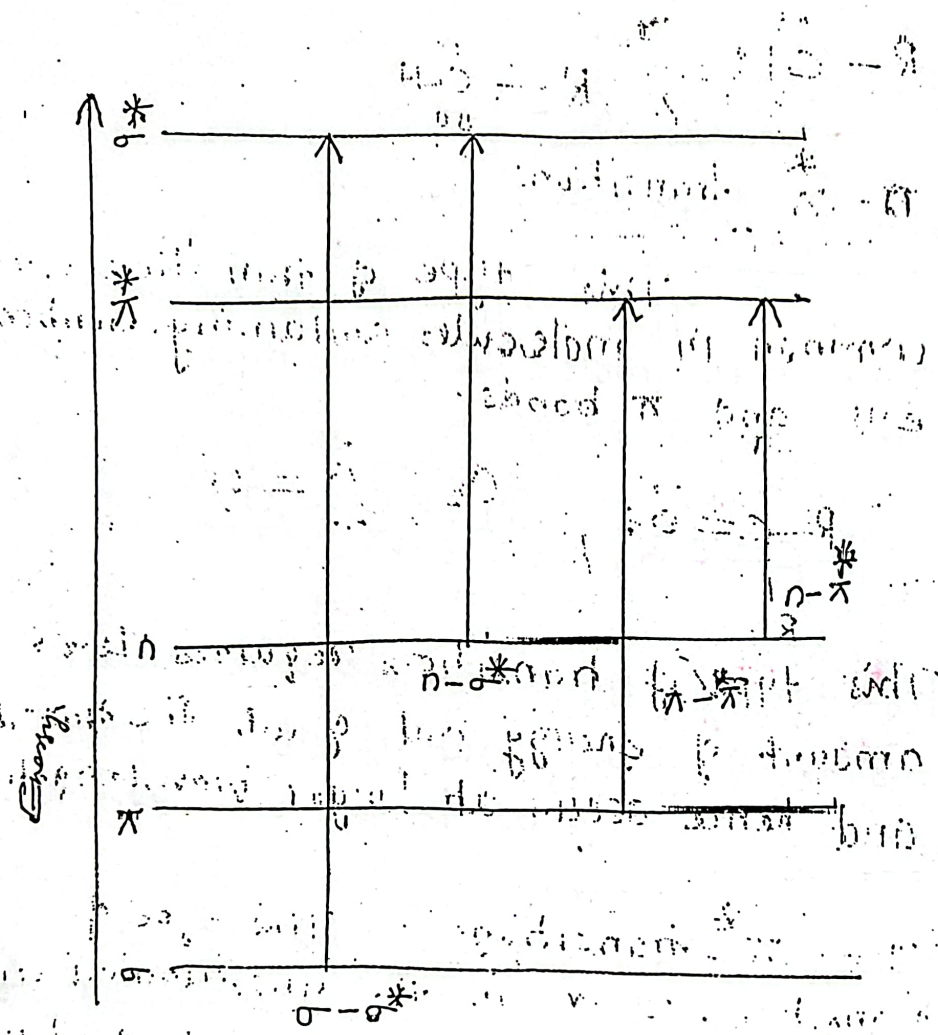
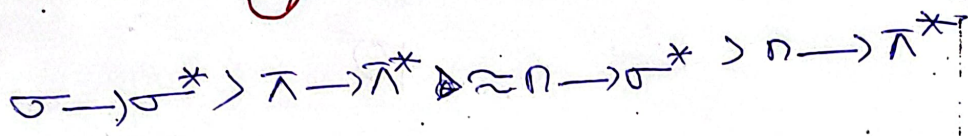
$\pi \rightarrow \pi$ bonding

$\pi^* \rightarrow \pi$ anti bonding

$n \rightarrow n$ non bonding

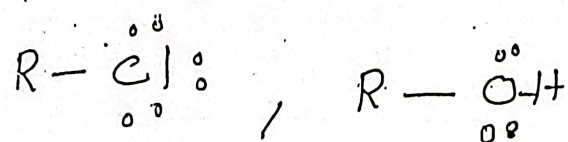
order of energy transition

9



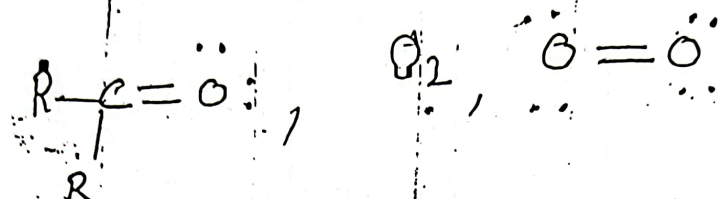
$\sigma \rightarrow \sigma^*$ transitions:- The energy required is larger. For saturated it is like methane, propane etc, absorption occurs near 150 nm (high energy). In methane, which has only four C-H σ bonds, and can have $\sigma \rightarrow \sigma^*$ transitions only, it shows λ_{max} in UV region.

$n \rightarrow \sigma^*$ transitions:- This type of transition takes place in saturated compounds containing one heteroatom with unshared pair of electrons. eg: R-X, R-Cl, R-OH,



$n \rightarrow \pi^*$ transitions:

This type of transitions are common in molecules containing nonbonding e^- s and π bonds.

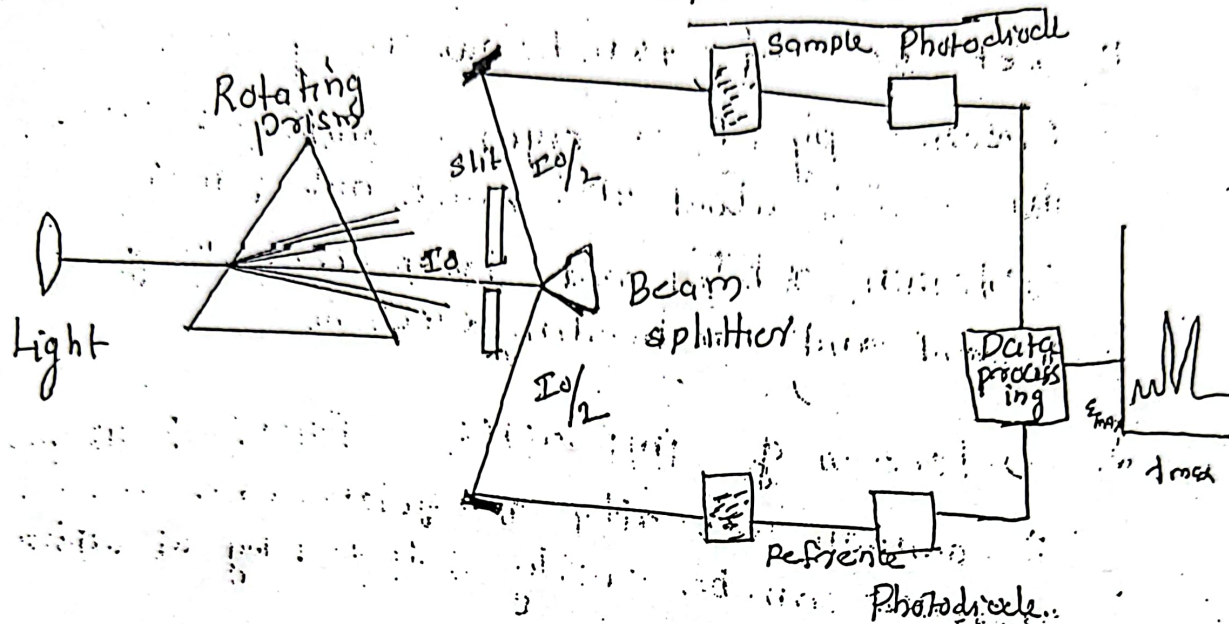


This type of transition requires least amount of energy out of all the transitions and hence occur at longer wavelengths.

$\pi \rightarrow \pi^*$ transitions :- This type of transition occurs in the unsaturated centers of the molecule i.e. in compounds containing double or triple bonds and also in aromatics e.g. alkenes, alkynes, cyanides etc. conjugated diene

Chromophores :- All those compounds which absorb light of wavelength between 400 - 800 nm appear coloured to the human eye. A chromophore was considered any system which is responsible for imparting colour to the compound. Nitro compounds are generally yellow in colour. Similarly aryl conjugated azo group is a chromophore.

Instrumentation of UV-visible Spectrophotometer



The prism is used to disperse the light into different wavelength. The desired wavelength is selected by rotating the prism for letting the selected wavelength to pass through the narrow slit. This beam is split into two equal halves of same intensity by means of a beam splitter. One of the beams then made to pass through a sample in a cell. Cell made of a material which is transparent to the wavelength under study. After passing through the cell, the light is made to follow on a photodiode which converts light to electric current. The output of the photocell (photodiode) which converts light to electric current. The output of the photocell depends on the intensity of light incident on the photocell. Similarly, the other beam is made to pass through identical reference cell of the same material and solvent. The output of reference cell is only converted to corresponding electric current by a second photocell. These two outputs then made to interfere to give the output as a spectrum.

Applications of UV-Visible Spectroscopy

1. Detection of functional group:- UV Spectroscopy of an organic compound gives information about the functional group containing π bond and groups containing π bond and non-bonding electrons.

2. Detection of impurities:- For eg: Benzene is common impurity in cyclohexane and its presence can be easily detected by absorption at 255 nm.

3. In quantitative analysis:- By using Beer Lambert's law, $A = \log \frac{I_0}{I} = \epsilon \cdot l \cdot c$

Where A is the absorbance, ϵ is the molar absorption coefficient, c is the concentration of the solution and l is the thickness of solution.

4. Determination of molecular weight
5. Study of equilibrium in solutions
6. Study of kinetics of chemical reaction
7. Study of kinetics of chemical reaction